CORA: Structure-Aware Active Causal Reorganization in Multi-Agent Reinforcement Learning

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Abstract

Multi-agent teams may need to adapt not only individual actions but also their organizational structure: roles, mission assignments, permissions, and delegation. The consequences of compound reorganizations can be delayed, nonlinear, and causally coupled through shared resources and adaptive policies. We formulate constrained causal organizational reconfiguration (CCOR) and specify CORA, a proposed neuro-symbolic approach integrating causal dependency structures, compositional effect estimation, decision-focused active intervention design, and conservative selection under symbolic constraints. We provide conditional results for interaction locality, coefficient and decision identifiability, estimation uncertainty, and expected-value non-degradation. These results build on established mathematics and are not presented as independently novel. We also specify CORA-Bench, a three-level, fully automated validation program, and four distinct simulation applications. No implementation results, empirical gains, or confirmed priority claims are asserted in this draft.

Draft status. Mathematical statements are conditional on explicit assumptions. Proposed algorithmic components, experimental predictions, and claims of novelty remain to be verified. No human-subject evaluation is required. The final manuscript must undergo bibliographic verification and venue-policy review.

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1 Introduction

MARL learns coordination policies for interacting autonomous agents. Organizational multi-agent systems additionally make coordination structures explicit through roles, missions, relationships, responsibilities, and normative constraints. When agents fail, missions appear, resources change, or team composition evolves, adapting only individual policies may not suffice: the organization itself may need to change.

The question is not simply whether one organization predicts a higher reward than another. It is what would happen under a specified intervention on the organization, with a defined continuation policy and a controlled pre-intervention context. The effects of two organizational edits need not add: a communication link may be useless without an observer, whereas two individually useful reallocations may compete for the same scarce resource. In large organizations, the number of possible compound edits is exponential.

Research problem. How can decentralized learning agents autonomously select effective organizational reconfigurations under changing conditions, using causal estimates of their consequences while preserving explicit organizational constraints?

Scope. We begin with cooperative partially observable multi-agent processes and a well-defined collective return. The theory is domain-independent. Proposed applications include warehouse robotics, search and rescue, wildfire response, and simulated hospital operations. Self-interested agents and strategic equilibrium are outside the initial formal guarantees.

Candidate contributions. (C1) A formal constrained causal organizational reconfiguration problem; (C2) a conservative intervention dependency model; (C3) decision-focused active experimentation with compositionally structured effects; (C4) confidence-aware organizational selection; (C5) a multi-level automated evaluation benchmark. These are intended contributions, not experimentally established findings.

2 Related Work and Novelty Boundaries

Organizational models have long represented roles and structures, including organizational adaptation [1]. Learning roles is also established in MARL (ROMA and RODE) [2, 3]; MAPPO is a relevant cooperative baseline [4]. Causal intervention semantics are classical [5], as are factorial interaction decompositions and sequential confidence bounds [6]. Thus none of the following can be claimed as new in isolation: organizational reconfiguration, role-conditioned policies, causal interventions, factorized effects, active experiment design, or conservative confidence-bound decisions.

A publication-ready comparison must examine full texts on organizational adaptation via MDPs, causal games, compositional causal-effect estimation, agent-directed norm synthesis, formal contracts, and targeted causal MARL interventions. The remaining potential gap is their *decision-focused coupling under explicit organizational constraints and adaptive policies*, provided that the resulting method is shown to outperform closely matched structured and active-design baselines. This novelty claim remains provisional.

3 Formal Problem

Let $\mathcal{N} = \{1, \ldots, n\}$ denote heterogeneous agents, s_t an environment state, a_t^i a local action, o_t^i an observation, and h_t^i the local history. The transition kernel is $P(s_{t+1} \mid s_t, a_t, G_t)$ and each decentralized policy is $\pi_i(a_t^i \mid h_t^i, \zeta_t^i)$, where ζ_t^i is its authorized organizational view.

An explicit organization is $G = (A, R, M, E, D, \rho)$: agents, roles, missions, typed relations, normative rules, and assignments. A validator $\mathrm{Valid}(G)$ checks structural and deontic invariants. Given elementary edits $\delta_1, \ldots, \delta_m$, a compound intervention is $z \in \{0,1\}^m$, with support $S(z) = \{j : z_j = 1\}$. The transaction operator $T_G(G, z)$ must define the outcome of concurrent edits, including conflicts. The admissible set is

$$\mathcal{Z}_{\mathrm{valid}}(G) = \{z \in \{0,1\}^m : \mathrm{Valid}(T_G(G, z)) = 1\}. \tag{1}$$

The no-op $z_0 = \mathbf{0}$ is always a candidate. Binary sets of edits are inappropriate when edit order matters unless an atomic ordering or conflict-resolution semantics is explicitly specified.

At a decision time, fix a pre-intervention information context $X = x$, horizon K, and continuation protocol Π (frozen policies or an explicitly specified learning update rule). Define potential outcome and intervention value:

$$Y^{(K)}(z; \Pi) = \sum_{k=0}^{K-1} \gamma^k r_{t+k}(z; \Pi), \qquad F_x^{\Pi}(z) = \mathbb{E}[Y^{(K)}(z; \Pi) \mid X = x]. \tag{2}$$

With reorganization cost $c_x(z)$, set $J_x^{\Pi}(z) = F_x^{\Pi}(z) - c_x(z)$. The ideal decision maximizes J over $\mathcal{Z}_{\mathrm{valid}}(G)$. This intervention value is not generally identified from passive correlations. Controlled randomized interventions, consistency, logged assignment probabilities, and sufficient positivity permit estimation on the experimentally supported intervention set. Simulator resets provide evaluation counterfactuals only if the simulator implements the required interventions faithfully.

4 Counterexamples and Exact Interaction Decomposition

Consider two edits with $F(\varnothing) = F(\{1\}) = F(\{2\}) = 0$ and $F(\{1,2\}) = 10$. The interaction is ten; an additive estimator predicts zero. Alternatively, two systems may agree on all singleton values $F(\varnothing) = 0$, $F(\{1\}) = F(\{2\}) = 1$ but have pair values 2 and -5. Singleton experiments cannot distinguish them.

Proposition 4.1 (Singleton non-identifiability)**.** *For $m \geq 2$, no-op and singleton intervention values do not determine arbitrary compound intervention values.*

Proof. Two set functions can agree on every subset of cardinality at most one but disagree on a subset of cardinality two. □

For any set function $F : 2^{[m]} \to \mathbb{R}$, the Möbius interaction coefficients are

$$\phi(T) = \sum_{B \subseteq T} (-1)^{|T|-|B|} F(B), \qquad F(S) = \sum_{T \subseteq S} \phi(T). \tag{3}$$

This is an exact classical identity, not a new theorem. In general there are 2^m coefficients. An organizational graph alone does not imply sparse interactions: two disconnected teams can compete through a shared physical resource or a global reward.

5 T1: Causal Locality

Let $\mathcal{D}_K$ be a correct time-unrolled acyclic structural causal graph containing intervention nodes U_j, transitions, observations, actions, organizational changes, reward factors, and policy updates when applicable. Suppose $Y^{(K)} = \sum_{\ell=1}^{L} Y_\ell$ and define $D_\ell = \{j : U_j \in \mathrm{Anc}_{\mathcal{D}_K}(Y_\ell)\}$.

Assumption 5.1 (Interventional ancestry)**.** *Conditional on a pre-intervention context x, the interventional expectation of each reward factor depends only on the intervention coordinates in D_ℓ. Interventions modify only designated structural mechanisms, and x contains no post-treatment conditioning.*

Theorem 5.2 (T1: locality of interactions)**.** *There exist functions $f_{\ell,x}$ such that*

$$F_x(S) = \sum_{\ell=1}^{L} f_{\ell,x}(S \cap D_\ell). \tag{4}$$

For nonempty T, if $T \not\subseteq D_\ell$ for all ℓ, then $\phi_x(T) = 0$.

Proof. Linearity of expectation and the ancestry assumption give Eq. (4). Substitute into Eq. (3). For each factor ℓ choose $j \in T \setminus D_\ell$. Pair every subset $B \subseteq T$ not containing j with $B \cup \{j\}$. Both have the same projection onto D_ℓ and opposite signs, so the corresponding terms cancel. □

Corollary 5.3. *If $|D_\ell| \le d$, the candidate interaction family $\mathcal{K} = \bigcup_{\ell=1}^{L} 2^{D_\ell}$ has cardinality at most $\min(2^m, \sum_\ell 2^{|D_\ell|}) \le \min(2^m, L2^d)$.*

Theorem 5.4 (Restricted reconstruction)**.** *Let $\mathcal{K}$ be downward closed and contain $\varnothing$. If $\phi(T) = 0$ outside $\mathcal{K}$, then $F(S) = \sum_{T \subseteq S, T \in \mathcal{K}} \phi(T)$, and each included coefficient is determined by $F(B)$ for $B \subseteq T$.*

Proof. Apply Möbius inversion and discard the zero terms; downward closure ensures every subset required by inversion remains in $\mathcal{K}$. □

Remark 5.5. A safe over-approximation of causal ancestry preserves correctness but can be dense. A graph missing actual dependencies introduces bias. If a required combination is forbidden, its value cannot be directly measured; reconstruction then needs further assumptions, a different estimand, or a decision-focused identifiability analysis.

6 T2: Identifiability and Estimation

For interaction family $\mathcal{K}$ let $p = |\mathcal{K}|$ and $\varphi_T(z) = \prod_{j \in T} z_j$. Under exact specification $F(z) = \varphi(z)^\top \theta$. Collect N interventions with feature matrix $X \in \mathbb{R}^{N \times p}$ and noisy outcomes $Y = X\theta + \epsilon$.

Theorem 6.1 (T2a: coefficient identifiability)**.** *The coefficient vector is identifiable from intervention means if and only if* $\operatorname{rank}(X) = p$.

Proof. The map $\theta \mapsto X\theta$ is injective exactly when $\operatorname{Null}(X) = \{0\}$. □

Proposition 6.2 (Decision-relevant identifiability)**.** *The value at target z is identifiable if and only if $\varphi(z) \in \operatorname{Row}(X)$.*

Proof. The target linear functional is constant on observationally equivalent parameter vectors precisely when it annihilates $\operatorname{Null}(X)$. □

Forbidden combinations may prevent coefficient identification without preventing identification of all admissible decisions.

Assumption 6.3 (Fixed-design estimation)**.** *The context and continuation policy are fixed, the design is fixed independently of future noise, errors are independent mean-zero σ-sub-Gaussian, and X has full column rank.*

Theorem 6.4 (T2b: prediction concentration)**.** *For least squares $\hat{\theta} = (X^\top X)^{-1} X^\top Y$ and a fixed target z, with probability at least $1 - \delta$,*

$$|\varphi(z)^\top(\hat{\theta} - \theta)| \le \sigma\sqrt{2\log(2/\delta)}\sqrt{\varphi(z)^\top (X^\top X)^{-1}\varphi(z)}. \tag{5}$$

Proof. The prediction error is a weighted sum of independent sub-Gaussian errors with squared weight norm $\varphi(z)^\top (X^\top X)^{-1}\varphi(z)$. Apply a two-sided concentration inequality. □

A union bound over M fixed targets replaces δ by δ/M. If $\lambda_{\min}(X^\top X) \ge N\kappa/p$, the radius is at most $\sigma\|\varphi(z)\|_2\sqrt{2p\log(2M/\delta)/(N\kappa)}$. This requires an informative design.

Adaptive experimentation. Fixed-design bounds are not directly valid for outcome-dependent experiment selection. For predictable features and conditionally sub-Gaussian martingale noise, use ridge matrix $V_N = \lambda I + \sum_{n\le N}\varphi(z_n)\varphi(z_n)^\top$, estimator $\hat{\theta}_N = V_N^{-1}\sum_{n\le N}\varphi(z_n)Y_n$, and the standard self-normalized bound

$$|\varphi(z)^\top(\hat{\theta}_N - \theta)| \le \beta_N\|\varphi(z)\|_{V_N^{-1}}, \quad \beta_N = \sigma\sqrt{2\log\frac{\det(V_N)^{1/2}}{\det(\lambda I)^{1/2}\delta}} + \sqrt{\lambda}S, \tag{6}$$

when $\|\theta\|_2 \le S$. This is established linear-bandit theory [6], not a novel CORA bound.

7 T3–T4: Reliable Reorganization

Let $\hat{F}(z)$ estimate a restricted interaction model under reference policy Π_0, while deployment uses Π_1. Distinguish

$$|\hat{F}(z) - F_x^{\mathcal{K},\Pi_0}(z)| \le \epsilon_{\text{stat}}(z), \tag{7}$$

$$|F_x^{\mathcal{K},\Pi_0}(z) - F_x^{\Pi_0}(z)| \le \epsilon_{\text{struct}}(z), \tag{8}$$

$$|F_x^{\Pi_0}(z) - F_x^{\Pi_1}(z)| \le \epsilon_{\text{shift}}(z). \tag{9}$$

The structural error is bounded by $\sum_{T\subseteq S(z), T\notin\mathcal{K}}|\phi(T)|$ when the full set function exists. This bound cannot be certified from a guessed sparse graph alone.

Theorem 7.1 (T4: conditional non-degradation)**.** *Suppose statistical bounds hold simultaneously for all admissible interventions with probability $1 - \delta$, structural and policy-shift bounds are valid, and intervention costs are known. Define $B(z) = \epsilon_{\text{stat}}(z) + \epsilon_{\text{struct}}(z) + \epsilon_{\text{shift}}(z)$ and $\hat{J}(z) = \hat{F}(z) - c(z)$. If CORA accepts z only when*

$$\hat{J}(z) - B(z) \ge \hat{J}(z_0) + B(z_0), \tag{10}$$

then $J_x^{\Pi_1}(z) \ge J_x^{\Pi_1}(z_0)$ with probability at least $1 - \delta$.

Proof. The triangle inequality yields $|\hat{F}(z) - F_x^{\Pi_1}(z)| \le B(z)$ on the coverage event. Thus $J(z) \ge \hat{J}(z) - B(z) \ge \hat{J}(z_0) + B(z_0) \ge J(z_0)$. □

Remark 7.2. This concerns expected finite-horizon values on a confidence event, not guaranteed safety of every realized trajectory. The argument is standard confidence-bound reasoning; useful bounds for misspecification and policy shift are the research challenge.

7.1 A policy-shift bound

If joint policies Π_0, Π_1 have total variation distance at most α at each common relevant history, share the same transitions, and $|r_t| \le R_{\max}$, maximal coupling gives

$$\epsilon_{\text{shift}}(z) \le 2R_{\max}\sum_{t=0}^{K-1}\gamma^t[1 - (1-\alpha)^{t+1}] \le 2R_{\max}\alpha\sum_{t=0}^{K-1}(t+1)\gamma^t. \tag{11}$$

For $\gamma < 1$ this is bounded by $2R_{\max}\alpha/(1-\gamma)^2$. It may be loose; importantly α must bound the joint policy, not merely each agent independently.

8 CORA: Candidate Algorithm

CORA combines a conservative causal dependency model, symbolic valid-edit generator, structured intervention estimator, decision-focused active experimentation, and confidence-aware commit rule. It must not assume that the symbolic organizational graph alone equals the dynamic causal graph.

8.1 Decision-focused acquisition

For two candidates let $d(z, z') = \varphi(z) - \varphi(z')$ and $V_N = \lambda I + \sum_{n=1}^{N} \varphi(z_n)\varphi(z_n)^\top$. An exploratory intervention a gives $V_{N+1}(a) = V_N + \varphi(a)\varphi(a)^\top$. A candidate criterion is

$$a^* \in \arg\min_{a \in \mathcal{Z}_{\text{valid}}} \left\{ \max_{z,z' \in \mathcal{C}_N} d(z, z')^\top V_{N+1}(a)^{-1} d(z, z') + \eta c_{\text{exp}}(a) \right\}, \tag{12}$$

where $\mathcal{C}_N$ is the set of plausibly optimal decisions. This resembles established optimal experimental design and best-arm identification; its mathematical form is not claimed new. Approximate combinatorial solvers must report optimization gaps and compute cost.

Algorithm 1 CORA research specification (not yet implemented)

Require: Simulator, initial valid organization, symbolic constraints, elementary edits, MARL learner, horizon K

Initialize organization-conditioned policies, causal structure, dataset and effect estimator
while training budget remains **do**
    Train behavioral policies; freeze continuation policy version
    Construct conservative temporal intervention dependencies and interaction features
    Generate admissible compound edits including no-op
    Select an informative valid experiment by decision-focused acquisition
    Execute controlled intervention branch and log context, assignment propensity, policy version, reward and cost
    Update causal effect estimator; audit design rank and confidence coverage
    Compare pessimistic intervention values to optimistic no-op value
    Atomically commit a certified valid edit, or retain organization
end while
return policies, organization selector, effect estimator, dependency graph, experiment logs

8.2 Distributed execution and experimental access

Decentralized action policies do not imply decentralized organizational commit. Concurrent proposals require conflict resolution, shared organization versioning, and atomic or consensus-backed adoption. Simulator branching for counterfactual training must not leak privileged information to deployed policies. Frozen-policy and continuously learning continuation protocols are distinct estimands.

9 Research Questions and Falsifiable Claims

RQ	Hypothesis	Decisive test / falsifier
RQ1	Structure-aware exploration identifies low-regret reorganizations with fewer interventions than uniform and structure-blind active design.	Regret against experiment budget; absence of gain against matched active design refutes advantage.
RQ2	Factorized models predict unseen compound interventions better when causal interactions are sparse and localized.	Held-out effect error by intervention cardinality and density; poor performance on sparse-local instances refutes the hypothesis.
RQ3	Valid uncertainty bounds reduce harmful expected-value reorganizations.	Harm rate, coverage, abstention and reward; no reduction or universal abstention undermines usefulness.
RQ4	A common algorithm improves adaptation across distinct simulated domains.	Transfer matrices, recovery curves, matched baselines; heavy domain-specific algorithm changes undermine generality.

10 CORA-Bench: Three-Level Experimental Design

10.1 Level 1: synthetic causal organizations

Generate organizations with m candidate edits and a ground-truth interaction family $\mathcal{K}$. Use $F(z) = \sum_{T \in \mathcal{K}} \theta_T \prod_{j \in T} z_j$ and $Y(z) = F(z) + \epsilon$, with controlled noise. Vary agent counts $4, 8, 16, 32, 64$; edit counts $4, 8, 16, 32, 64$; maximum interaction order 1–4; sparse-to-dense hypergraphs; intervention feasibility; noise; exact, conservative, and incomplete dependency graphs; and nonstationarity. Ground truth must remain hidden from learned methods. The constraint generator must audit actual feasibility, not assume a nominal constraint density determines it.

10.2 Level 2: sequential organizational dynamics

Simulate heterogeneous agents with capabilities, local observations, mission prerequisites and durations, resource contention, stochastic transitions, and organizational interventions. Maintain distinct organizational, environmental, and causal-dependency graphs. Obtain oracle effect estimates using controlled simulator branches with adequate Monte Carlo precision. Common random numbers can reduce variance but do not establish causal identification by themselves.

10.3 Level 3: adaptive MARL

Integrate organization-conditioned IPPO, MAPPO, and, when compatible, a value-factorization family. Evaluate frozen-policy and adaptive-policy continuation separately, preserving policy version metadata. Perturb team composition, tasks, resources, communication and observation quality. Enforce equal environment interaction budgets and comparable tuning budgets.

10.4 Baselines and ablations

Method	Question answered
Uniform/random intervention exploration	Is active design beneficial?
Global linear and neural effect estimators	Does causal factorization help beyond model capacity?

Method	Question answered
Factorized estimator with uniform design	Does decision-focused exploration add value?
Structure-blind active design	Does organizational structure add value?
Published compositional causal estimator	Does CORA outperform the closest causal modeling work?
Published organizational adaptation method	Does CORA outperform reorganization planning baselines?
CORA without confidence bounds	Does conservative selection prevent damaging interventions?
CORA with incomplete graph	How robust is the structural assumption?
Exact or high-precision simulator oracle	How much decision regret remains?

10.5 Metrics

Causal MAE on held-out admissible interventions is $|\mathcal{Z}_{\text{test}}|^{-1} \sum_{z \in \mathcal{Z}_{\text{test}}} |\hat{F}(z) - F(z)|$. Simple decision regret is $J(z^*) - J(\hat{z})$. Sample efficiency is the number of experiments required to reach a preregistered regret threshold. Harmful expected-value reorganization rate is the fraction of executed non-noop interventions with $J(z) < J(z_0)$; this must be distinguished from the frequency of unfavorable individual rollouts. Also measure constraint violations, abstention, computation, recovery time, calibration, confidence coverage, and performance under policy shift.

10.6 Statistics and reproducibility

Use multiple independent seeds (ten or more when feasible), matched budgets, IQM and bootstrap confidence intervals, performance profiles, and appropriate multiple-comparison corrections. Freeze primary hypotheses and metrics before final experiments. Record software and hardware versions, hyperparameters, all seeds, wall-clock time, total compute and unsuccessful trials. Never fabricate measurements or select only visually favorable episodes.

11 Multi-Domain Simulated Applications

Warehouse logistics. Heterogeneous transport, inspection and loading robots face congestion, failures and changing orders. Interventions modify delegation, role assignment and mission responsibility. Measure throughput, task lateness, congestion, recovery and intervention cost. Render synchronized spatial trajectories and assignment overlays.

Search and rescue. Exploration, communication, detection and evacuation agents respond to blocked routes, newly detected victims and communication failures. Measure rescues, rescue latency, coverage, safety constraints and reorganization cost. Render responsibility changes and route evolution.

Wildfire response. Surveillance drones, response units and support agents respond to spreading fires, changing wind, unit loss and resource depletion. Measure protected area, response time, coverage, resource use and coordination. The ecological domain is an application of the generic contribution, not a restriction on the research problem.

Simulated hospital operations. Agents with different operational competencies manage patient queues, priority constraints, equipment and staffing changes. Measure waiting time, throughput, priority compliance, overload and recovery. This is an abstract simulation, not a claim of real clinical efficacy.

Transfer. Keep the same core estimator, experiment design and selector across domains. Distinguish new configurations within a domain from true cross-domain transfer; specify whether learned weights, organizational structures, or only algorithmic mechanisms transfer.

12 Figure and Visualization Plan

1. **Figure 1: Problem and method.** Before/after organizational graph and causal dependencies; demonstrate why additive edit scoring can fail.
2. **Figure 2: True vs inferred interaction structure.** Edge/hyperedge precision, recall, false omissions, and symbolic graph versus temporal causal graph.
3. **Figure 3: Sample efficiency.** Effect MAE and decision regret versus intervention budget with uncertainty bands.
4. **Figure 4: Structural phase diagram.** Performance gain over baselines across interaction order, graph density and misspecification.
5. **Figure 5: Reliability and adaptation.** Harm rate, abstention, return and recovery time after matched disruptions.
6. **Figure 6: Cross-domain generalization.** Train/test transfer heatmap and side-by-side visual rollouts.

Every figure must be generated from saved raw data by versioned scripts. A qualitative replay is supplementary evidence, not a replacement for aggregate quantitative evaluation.

13 Threats to Validity and Limitations

Construct validity. A symbolic organizational relation need not be a causal relation. High return does not establish interpretability, and compliance with rules does not guarantee desirable real-world outcomes.

Internal validity. The intervention oracle may leak privileged state; unequal tuning or rollout budgets may bias comparisons. Simultaneously adapting MARL policies can invalidate fixed-policy causal estimates.

External validity. Synthetic factorization may be unrealistically favorable. Complex environmental couplings can make the causal hypergraph dense. Multiple scenarios of one simulator do not automatically constitute independent domains.

Theory limitations. T1 assumes a correct temporal SCM and reward-factor ancestry; T2 assumes correctly specified features and identifiable designs; T4 depends on valid bounds, including structural and shift components. None establishes pathwise safety or unrestricted zero-shot causal transport.

Algorithmic limitations. Active design over combinatorial admissible interventions can be computationally intractable. Conservative confidence tests can lead to frequent abstention. Learning the dependency graph is a separate identification problem.

14 Planned Paper Contributions and Evidence Traceability

ID	Intended contribution	Mathematical basis	Required evidence
C1	Formal CCOR semantics	Definitions and causal estimand	Unit tests and intervention consistency
C2	Structured effect estimation	T1, Möbius, T2	Unseen composition MAE and scaling
C3	Decision-focused exploration	Acquisition objective	Regret and sample efficiency against active design
C4	Conservative reorganization	T4 conditional rule	Confidence coverage, harm rate, abstention
C5	Cross-domain automated benchmark	Three-level protocol	Baselines, ablations, generalization and replays

15 Discussion and Future Work

The central unresolved scientific question is whether organizational causal dependencies can be inferred or conservatively approximated tightly enough to provide useful factorization, and whether decision-focused active experiments improve decisions over existing optimal-design and compositional causal methods. Future extensions include simultaneous decentralized proposal and consensus protocols, strategic agents, dynamic institution formation, learned graph uncertainty, and longer-horizon policy adaptation. These are not claimed to have been solved.

16 Conclusion

This draft formalizes causal organizational reconfiguration and specifies CORA as a testable algorithmic hypothesis. The strongest established mathematical results are conditional locality, identifiability, confidence concentration and conservative expected-value comparison. The prospective research contribution lies in exploiting organizational causal structure to select informative interventions and reliable reorganizations under constraints. A publication claim will require direct evidence against close prior methods, across synthetic, sequential and adaptive MARL settings.

References

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[6] Y. Abbasi-Yadkori, D. Pál, and C. Szepesvári. Improved Algorithms for Linear Stochastic Bandits. *NeurIPS*, 2011.

A Implementation Invariants

Tests must verify exact Möbius reconstruction on small full factorial designs, non-identifiability under rank-deficient designs, correct handling of forbidden interventions, atomic transaction validity, confidence coverage under specified noise assumptions, and no access to oracle values from learning modules. Counterexample tests should deliberately introduce omitted causal interactions and changing continuation policies.

B Open Specification Decisions

Before coding, fix: (i) the exact temporal SCM and observable organizational context; (ii) whether edits commute and how transactions commit; (iii) the admissible intervention language; (iv) whether ground-truth factors are known only to the benchmark oracle; (v) confidence calibration under misspecification; (vi) the policy versioning protocol; (vii) the comparator implementations and budgets; and (viii) whether the paper claims decentralized proposals, decentralized consensus, or only decentralized action execution.